ModelIt!

Student Activity Pack

The Brain-Machine Connection

How Scientists Are Learning to Read Your Mind

Grade NE

NGSS MS-LS1-8, HS-LS1-2, HS-ETS1-2, HS-ETS1-4

Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In the Level 1 Telephone Game lesson, you modeled how signal strength, distance, and noise affect message accuracy. What do you think happens when we add MORE sensors to capture the signal?**

2. **If a person who lost their arm could control a robotic hand with their thoughts, what challenges do you think the technology would face?**

3. **What does 'practice makes perfect' mean in terms of your brain and body learning to work together?**

4. **Draw what you think happens between a person's brain and a robotic arm — what are the steps?**

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Stimulus Intensity | Electrode Count | Noise Level | Signal Strength | Encoding Accuracy | Response Time | Motor Output Precision | Adaptation Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Best Case

Strong stimulus, many electrodes, quiet environment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Worst Case

Weak stimulus, few electrodes, noisy environment

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Adaptation Test

Optimal setup over multiple trials

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Feedback Loop — Information sent BACK from the robotic limb to the brain — does the user feel what the robot touches?

Algorithm Complexity — How sophisticated the computer software is at filtering noise and decoding signals — more complex algorithms take longer but are more accurate

Electrode Invasiveness — Whether sensors sit on the scalp (safe but weak signals) or are implanted in the brain (risky but strong signals)

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How can scientists decode the electrical language of your brain well enough to move a robotic arm — and why is it so incredibly hard? Use evidence from your simulation to support your answers.

1. **How is the Level 2 model different from the Level 1 Telephone Game model? What does adding more components reveal about BCI systems?**

2. **Trace the full pathway: Stimulus → Signal Strength → Encoding Accuracy → Motor Output Precision. What happens at each step when noise increases?**

3. **Why is adaptation such a powerful factor in BCI performance? What does it tell us about the relationship between brains and machines?**

4. **Which scenario surprised you the most and why?**

5. **If you were designing a BCI for a real patient, which component would you try to optimize first — and why?**

8 STEM Challenge: Prosthetic Hand Design

ModelIt!

YOUR ENGINEERING MISSION

Design and build a cardboard-and-string mechanical hand that a partner controls through a barrier using only coded signals — then test how noise and practice affect performance.

THE SCENARIO

A biomedical engineering lab needs a demonstration showing why brain-computer interfaces are so challenging. Your team will build a mechanical hand and a signal system that shows how encoding, noise, and adaptation affect motor control — just like a real BCI.

Engineering Design Thinking

1. HOW WILL YOU DESIGN THE CODING SYSTEM — HOW MANY DISTINCT SIGNALS DO YOU NEED FOR FIVE FINGERS?
2. WHAT MATERIALS WILL GIVE YOUR HAND THE MOST REALISTIC FINGER MOVEMENT?
3. HOW WILL YOU TEST THE EFFECT OF NOISE ON YOUR SIGNAL ACCURACY?
4. HOW WILL YOU MEASURE WHETHER ADAPTATION (PRACTICE) IMPROVES PERFORMANCE OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Neuroengineers design brain-computer interfaces that help paralyzed patients control robotic limbs, allow locked-in patients to communicate, and treat neurological disorders like epilepsy and Parkinson's

disease. They work at research hospitals, tech companies, and universities. They earn $85,000–$150,000/year.